Supplementary Material S2
MMAT Appraisal by Study and Domain

Facial Emotion Recognition through Machine Learning in Children with Autism Spectrum Disorder: A systematic scoping review of open-access computational evidence and of the gaps toward therapeutic monitoring

This document complements the figure in the manuscript presenting the aggregate distribution of MMAT judgements by domain (Section 2.6). Here the appraisal is reported study by study. In accordance with the manual of the Mixed Methods Appraisal Tool (MMAT) version 2018, which deliberately eliminated the numerical summative scoring present in earlier versions, no aggregate score per study is computed or reported. Each study was assigned to the corresponding design category and appraised with the five criteria proper to that category: criteria 3.1–3.5 for quantitative nonrandomised studies (Cat. 3), criteria 4.1–4.5 for quantitative descriptive studies (Cat. 4), and criteria 5.1–5.5 for mixed-methods studies (Cat. 5).

The five judgement columns correspond to the five aggregated domains described in Section 2.6 of the manuscript: **Rep.** = Representativeness of the sample (3.1/4.2); **Meas.** = Adequacy of measurements (3.2/4.3); **Int.** = Intervention as intended and sampling strategy (3.5/4.1); **Conf.** = Control of confounding factors and analysis (3.4/4.5); **Out.** = Completeness of outcome data (3.3/4.4). Judgements are coded as **Y** = Yes, **N** = No, **ND** = Not Determinable.

Section A. Eligible corpus (21 studies)

Of the 21 eligible studies, 20 are assignable to an MMAT design category (16 quantitative nonrandomised, 3 quantitative descriptive, 1 mixed methods) and 1 falls outside the instrument's scope (systems design without supervised classification). The sum $16+3+1+1$ recovers the 21 studies of the eligible corpus. Ahmad et al. (2024) was retracted on 17 August 2026 (*Healthc Technol Lett* 2026;13(1):e70095) and is excluded from this appraisal. Alsubai (2025) and Akın-Bülbül et al. (2024) are documented as contextual evidence in Section B.

#	Reference	Cat.	Rep.	Meas.	Int.	Conf.	Out.
Category 3: Quantitative nonrandomised (16 studies)							
1	Johnson et al. (2025)	3	Y	Y	Y	ND	Y
5	Shaik et al. (2025)	3	ND	Y	ND	N	Y
8	Radočaj & Martinović (2025)	3	Y	Y	Y	Y	Y
9	Washington et al. (2022)	3	Y	Y	Y	Y	Y
10	Sholikah et al. (2024)	3	ND	Y	ND	N	Y
12	Sariyanidi et al. (2025)	3	Y	Y	Y	Y	Y
13	Hassan et al. (2025)	3	Y	Y	Y	Y	Y
15	Rudovic et al. (2018)	3	Y	Y	Y	Y	Y
16	Cicceri et al. (2026)	3	Y	Y	Y	ND	Y
17	Rudovic et al. (2019)	3	Y	Y	Y	Y	Y
19	Alam et al. (2025)	3	ND	Y	ND	N	Y

(Continues on next page)

(Continued from previous page)

#	Reference	Cat.	Rep.	Meas.	Int.	Conf.	Out.
20	Hwang et al. (2024)	3	Y	Y	Y	Y	Y
21	Aljbori et al. (2025)	3	N	Y	N	N	Y
23	Vedantham et al. (2025)	3	N	Y	ND	N	Y
24	Hosney et al. (2024)	3	N	Y	N	N	Y
28	Leo et al. (2018)	3	Y	Y	Y	Y	Y
Category 4: Quantitative descriptive (3 studies)							
2	He et al. (2019)	4	Y	Y	Y	Y	Y
3	Reed et al. (2021)	4	Y	Y	Y	Y	Y
27	Dapogny et al. (2019)	4	Y	Y	Y	Y	Y
Category 5: Mixed methods (1 study)							
4	Lecciso et al. (2021)	5	Y	Y	Y	ND	Y
Outside MMAT scope (1 study)							
18	Almeida et al. (2019)	Systems design without supervised classification. Quality discussed narratively in Section 6.2 of the manuscript.					

Section B. Contextual evidence outside the eligible set (9 studies)

The following nine studies do not form part of the eligible corpus and do not participate in any denominator of the manuscript. Five infringe CE(3) (exclusively non-facial data); three infringe the age criterion (adult participants); one infringes CI(3) (journal not indexed in SJR in the year of publication).

#	Reference	Reason	Cat.	Rep.	Meas.	Int.	Conf.	Out.
Category 3: Quantitative nonrandomised (5 studies)								
C1	El Rhatassi et al. (2024)	Adult participants	3	ND	Y	ND	N	Y
C2	El Rhatassi et al. (2025)	Adult participants	3	ND	Y	ND	N	Y
C4	Nagae & Lee (2022)	EDA only (CE3)	3	Y	Y	ND	ND	Y
C7	O'Sullivan et al. (2023)	Speech only (CE3)	3	Y	Y	Y	ND	Y
C8	Alsubai (2025)	Tabular, text and clinical records only (CE3)	3	N	Y	N	N	Y
Category 4: Quantitative descriptive (3 studies)								
C5	Papaioannou et al. (2021)	EEG only (CE3)	4	Y	Y	Y	ND	Y
C6	Xu & Ju (2023)	sMRI only (CE3)	4	Y	Y	Y	Y	Y
C9	Akın-Bülbül et al. (2024)	Journal not indexed in SJR (CI3)	4	Y	Y	Y	ND	Y

(Continues on next page)

(Continued from previous page)

#	Reference	Reason	Cat.	Rep.	Meas.	Int.	Conf.	Out.
Outside MMAT scope (1 study)								
C3	Levy et al. (2024)	Systems design; adult participants.						

Notes.

1. The four retracted studies (Güngör et al., 2025; Mahmood et al., 2025; Kumar et al., 2024; Ahmad et al., 2024) are excluded from the corpus and do not receive MMAT appraisal. The internal # column follows the corpus order shared with Supplementary Material S4 (C prefix for contextual evidence) and does not correspond to bibliographic reference numbers. Gaps in the sequence are not numbering errors: numbers were assigned before the final integrity and eligibility verification and are not reused.
2. Studies classified as systems design (Almeida et al., 2019; Levy et al., 2024) do not correspond to any MMAT design category; quality is discussed narratively in Section 6.2 of the manuscript.
3. ND in Conf. for Johnson et al. (2025), Lecciso et al. (2021) and Cicceri et al. (2026) reflects partial documentation of confounding control.
4. Negative judgements in Rep. and Conf. for Aljbori et al. (2025), Vedantham et al. (2025) and Hosney et al. (2024) are consistent with use of web repository datasets without clinical validation (Table 18 of the manuscript).
5. The MMAT does not capture risks specific to predictive modelling such as data leakage, image-level versus subject-level partitioning, or model calibration. This limitation is documented in Section 2.9 of the manuscript.